

IIT JODHPUR · VISUALIZATION AND STORYTELLING WITH DATA

# Climate Change: A Data-Driven Visual Story

Visualizing how human activity is reshaping Earth's climate — through data, design, and evidence.

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# Rising Temperatures, Rising CO<sub>2</sub>

+1.2°C

Global Warming

Above pre-industrial baseline

422 ppm

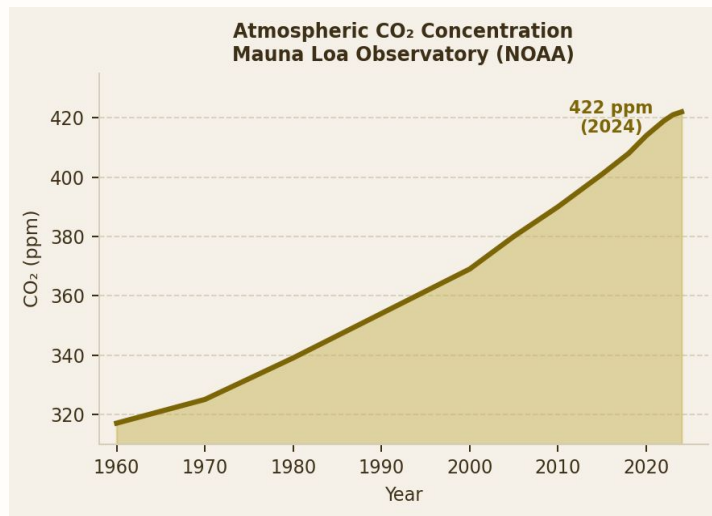
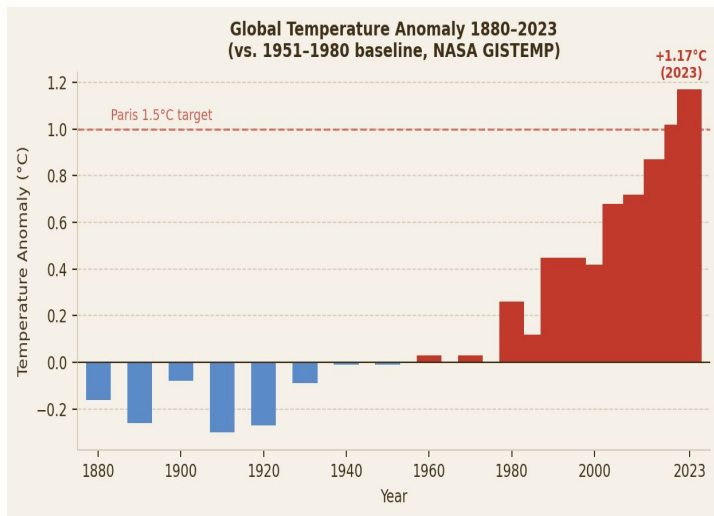
Atmospheric CO<sub>2</sub>

Highest in 800,000 years

21 cm

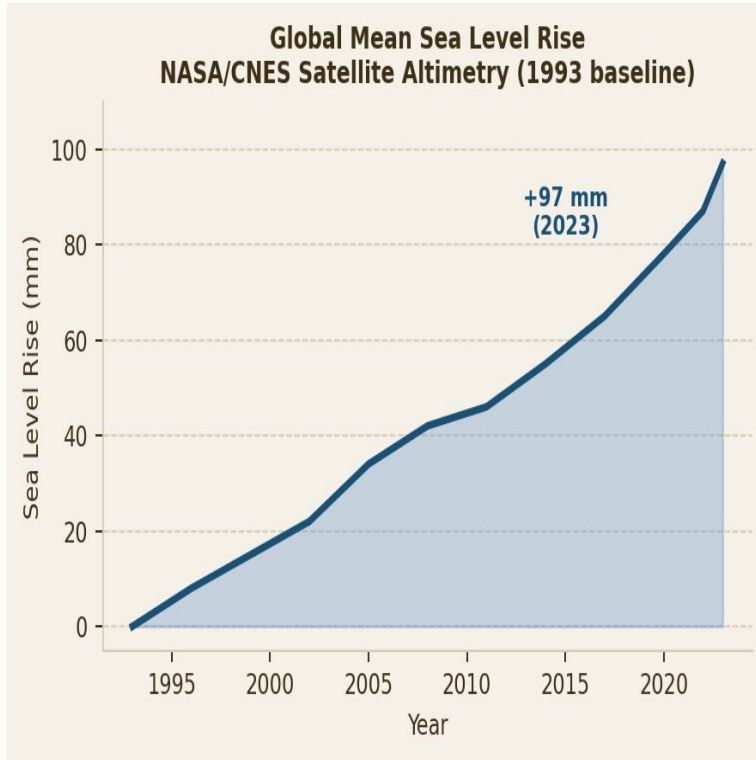
Sea Level Rise

Since 1880



2023 was among the hottest years recorded. CO<sub>2</sub> levels have accelerated sharply since industrialization, crossing 422 ppm — a concentration not seen in 800,000 years.

# Climate Instability Is Increasing Global Risk

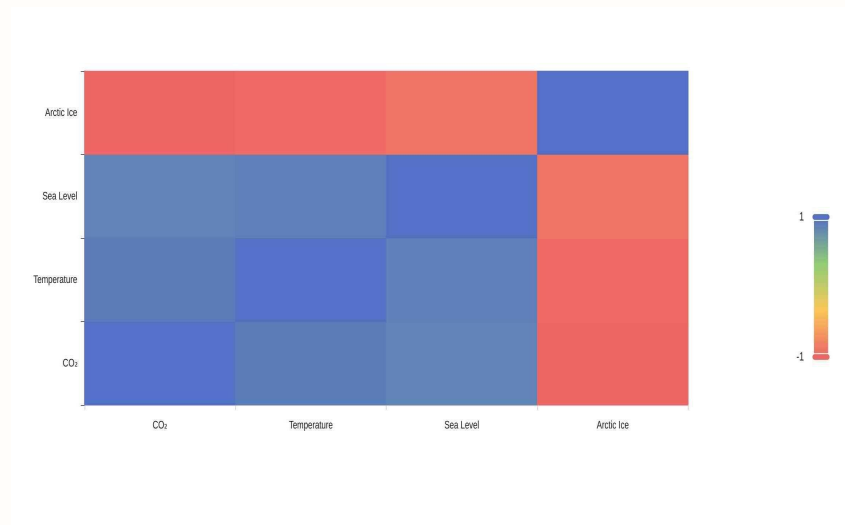


## Three Critical Threats

- **Sea Level Rise** — 21 cm since 1880, accelerating coastal flooding and erosion
- **Arctic Ice Melt** — Sea ice declining 13% per decade, amplifying heat absorption
- **Extreme Weather** — Category 4–5 hurricanes doubled in frequency since 1980

📌 These impacts are interconnected — each accelerates the others in a dangerous feedback loop.

# The Correlation Is Unmistakable



## What the Data Reveals

Greenhouse gas concentration **strongly correlates** with rising temperatures and environmental disruption across all measured variables.

CO<sub>2</sub> ↔ Temperature

Correlation: **0.91**

Temperature ↔ Sea Level

Correlation: **0.89**

CO<sub>2</sub> ↔ Arctic Ice

Correlation: **-0.88** (inverse)

## CONCLUSION

# Data Tells the Story. Action Writes the Future.



## Rising Temperatures

+1.2°C and accelerating



## Increasing Emissions

422 ppm CO<sub>2</sub> — record high



## Stronger Disasters

Extreme weather intensifying globally

Data visualization transforms climate data into compelling stories that drive awareness and action.

